**A taste of space**
The chinese goverment is commercialising the food that their "taikonauts" enjoy in space
**http://bit.ly/xBkA85**

**A space odyssey**
Arthur C. Clarke's most famous works. The book was written even before the launch of the moon missions but was acclaimed for being uncannily accurate

Even if we developed a ship that could travel at a tenth of the speed of light, it would still take us half a century to reach our nearest star. What if there is nothing there, and we want to come back? That would be a century of space travel, unless we can develop systems for hibernation, we would need to send a family or two on that trip. A family that might never see our solar system again, let alone the Earth. The need for vast quantities of supplies and a self-sufficient ship are paramount.

## The solution

Scientists are still unsure how to manage these things at a cosmic scale. The ship will have to be built in space over a period spanning several years, perhaps decades. There has been talk of launching unmanned cargo ships in advance of the ship while it is being constructed. These robotic ships would head along the course that the ship intends

The venerable Voyager I probe

to take and would have plenty of time to pick up speed without expending much fuel by the utilisation of unconventional means such as solar-powered ion thrusters. This makes a lot of sense as a spacecraft with a large mass is much harder to get off the line. Arthur C. Clarke didn't stray very far from the truth when he spoke of replenishment bases on Europa and similar moons. Such stops would be absolutely essential not just for refuelling, but for water supplies as well.

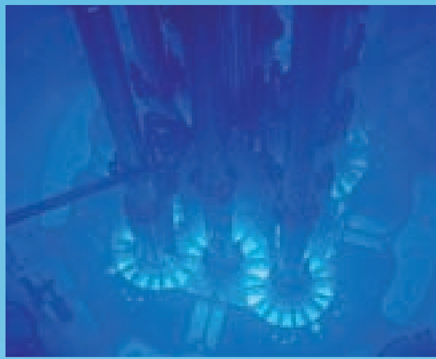
Food would have to be something that can be grown and replenished. The ship, self-sustaining in every possible way, a veritable Noah's ark with its own menagerie. Growing edible plants on the ship is completely out of the question, almost all food crops require sunlight and the spectrum of light provided by indoor lamps is simply not enough. The power required to drive "daylight" lamps required for such a job would be tremendous, not to mention the supply of water that would be required and, the spare lamps that will have to be carried. The only possible solution is freeze-dried foods that would have to be stored at cryogenic temperatures.

Once it leaves the solar system the crew will be subject to such loneliness as they would never have experienced before. Radio transmissions from the ship would not be able to reach the Earth, not just for lack of power, but also because the target would be so small that it would be almost impossible to hit as the distance keeps increasing. Even the space around would not have anything in it, nothing more than maybe a speck of dust per million cubic miles. The light from stars would provide no warmth and no power, it would be the same as trying to warm yourself by the light of the stars, only much colder, the ship, nothing more than an unmanageable hulk hurtling through space at more than 3,000 km/s; and despite that incredible speed, would still seem stationary with respect to its surroundings.

The only source of power in such a situation will have to be some sort of a sustained nuclear reaction. No other fuel will be compact or efficient enough to provide power for such a time period. In fact, nuclear power will be everything. The true heart of the ship, pumping its radioactive life blood throughout the ship just to generate heat.

## A monumental task

The ship so designed would have to be faster than any ship ever made, not to mention so large that it would probably end up like a miniature moon in itself.

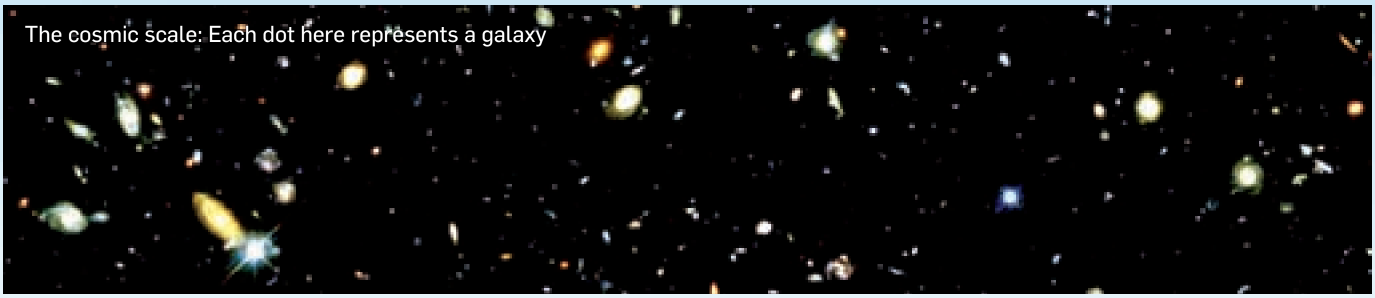


The glow of the future- A nuclear reactor

Unless there is a real possibility for hibernation, such a ship would be most impractical for any purpose other than some sort of emergency. With most of the current consumer electronics products lasting barely 3 years on average, a ship designed to function perfectly through a hundred years of the harshest conditions known to man will have to be something really extraordinary.

As it stands, such a ship is possible to construct, but at a cost that would be astronomical, considering that the ship is meant for space, that is understandable. Despite all this, the probability of failure during interstellar travel are extremely high.

Despite these challenges, once man sets his sight on inter-planetary travel, such ships would be absolutely essential. Behemoths, large enough to be miniature moons by themselves would end up as the taxis of the cosmos, ferrying passengers and cargo through the depths of space.



The cosmic scale: Each dot here represents a galaxy